Lecture 1 (September 2nd)

Definition. (Contour integral) Let C be a contour parametrised by $z = z(t)$ where $a \leq t \leq b$. Also, let f be continuous on C. Then, the contour integral of f is defined as

$$\int_C f(z)\,dz = \int_C u\,dx - v\,dy + i\int_C u\,dy + v\,dx$$

and also, through parametrisation, we have

$$\int_C f(z)\,dz = \int_a^b f(z(t))z'(t)\,dt$$

Definition. (Derivative) The derivative of a complex function is defined as

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

if it exists. If f is C^1 (the function is continuous up its first derivatives) and $u_x = v_y$ & $v_x = -u_y$, then $f'(z)$ exists.

Theorem. (Green's theorem) Let P and Q be a C^1 function on $\bar{D}$ where D is a simply connnected bounded region whose boundary is C. Then,

$$\int_C P\,dx + Q\,dy = \iint_D \Big(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\Big)\,dxdy$$

where C is oriented in the positive sense. Here,

$$\int_C P\,dx = -\iint_D \frac{\partial Q}{\partial y}\,dydx$$

and

$$\int_C Q\,dy = \iint_D \frac{\partial Q}{\partial x}\,dxdy$$

Proof. Notice that for a parametrised simple disc where the above is $y = g_2(x)$ and $y = g_1(x)$ on the bottom,

$$\int_C P\,dx = \int_a^b P(x, g_1(x))\,dx - \int_a^b P(x, g_2(x))\,dx$$

and

$$\iint_D \frac{\partial P}{\partial y}\,dydx = \int_a^b \int_{g_1(x)}^{g_2(x)} \frac{\partial P}{\partial y}\,dydx = \int_a^b \Big[P(x, g_2(x)) - P(x, g_1(x))\Big]\,dx$$

Similarly,

$$\int_D Q\,dy = \int_c^d Q(y, h_2(x))\,dy - \int_c^d Q(h_1(y), y)\,dy$$

and

$$\iint \frac{\partial Q}{\partial x}\,dxdy = \int_c^d \int_c^d \int_{h_1(y)}^{h_2(y)} \frac{\partial Q}{\partial x}\,dxdy = \Big[Q(h_2(y), y) - Q(h_1(y), y)\Big]\,dy$$

□

Theorem. (Cauchy's theorem) If f is a analytic (differentiable) inside and on a bounded simply connected domain D whose boundary is C and $f \in C^1(\bar{D})$ then

$$\int_C f(z)\,dz = 0$$

Proof. Let $f = u + iv$. Then $u_x = v_y$ and $v_x = -u_y$ on D. Thus

$$\int_C f(z)\,dz = \int_C u\,dx - v\,dy + i\Big[\int_C u\,dy + v\,dx\Big]$$

where

$$\int_C u\,dx - v\,dy = \iint_D \Big(-\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\Big)\,dxdy$$

and

$$\int_C u\,dy + v\,dx = \iint_D \Big(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\Big)\,dxdy$$

notice how the terms in the large brackets vanish, and that the integral itself equals zero. □

Theorem. (Generalised Cauchy's Theorem) If $f \in C^1(\bar{D})$ is analytic in a bounded multiply connected domain D with finitely many holes inside, then

$$\int_{\partial D} f(z)\,dz = 0$$

where ∂D is a positively oriented boundary of D.

Theorem. (Cauchy Integral Formula) If f is analytic inside and on a POSCC C then for every z_0 inside C we have

$$f(z_0) = \frac{1}{2\pi i}\int_C \frac{f(z)}{z - z_0}\,dz$$

Lecture 2 (September 4th)

Recall. (Cauchy's formula) For a positively oriented simple closed curve (POSCC) C and f that is analytic inside and on C, we have

$$f(z) = \frac{1}{2\pi i}\int_C \frac{f(\xi)}{\xi - z}\,d\xi$$

for all z inside C. If $\bar{D}(z_0, r) = \{z \,|\, |z - z_0| \le r\} \subset D$,

$$\frac{1}{\xi - z} = \frac{1}{\xi - z_0 + z_0 - z} = \frac{1}{(\xi - z_0)\Big(1 - \dfrac{z - z_0}{\xi - z_0}\Big)}$$

for $z \in D(z_0, r)$. However,

$$\Big|\frac{z - z_0}{\xi - z_0}\Big| < \frac{|z - z_0|}{r} < 1$$

so that

$$\frac{1}{\xi - z} = \frac{1}{\xi - z_0}\sum_{n=0}^{\infty}\Big(\frac{z - z_0}{\xi - z_0}\Big)^n = \sum_{n=0}^{\infty}\frac{(z - z_0)^n}{(\xi - z_0)^{n+1}}$$

Thus

$$\begin{aligned} f(z) =& \frac{1}{2\pi i}\int_C f(\xi)\sum_{n=0}^{\infty}\frac{(z - z_0)^n}{(\xi - z_0)^{n+1}}\,d\xi \\ =& \sum_{n=0}^{\infty} a_n (z - z_0)^n \end{aligned}$$

where

$$a_n = \frac{1}{2\pi i}\int_C \frac{f(\xi)}{(\xi - z_0)^{n+1}}\,d\xi = \frac{1}{n!}f^{(n)}(z_0)$$

Notice that the convergence is uniform for $\xi \in C$.

Theorem. If f_n is continuous in C such that $f_n \to f$ uniformly on C

$$\int_C f(z)\,dz = \lim_{n\to\infty}\int_C f_n(z)\,dz$$

Theorem. If $f(z) = \sum_{n=0}^{\infty} c_n (z - z_0)^n$ for $z \in D(z_0, R)$ then for $0 < r < R$ we have

$$\frac{1}{2\pi}\int_0^{2\pi} |f(z_0 + re^{it})|^2\,dt = \sum_{n=0}^{\infty} |c_n|^2 r^{2n}$$

Proof. Take

$$f(z_0 + re^{it}) = \sum_{n=0}^{\infty} c_n r^n e^{int}$$

so that

$$
\begin{aligned}
\frac{1}{2\pi}\int_0^{2\pi}|f(z_0+re^{it})|^2\,dt =& \frac{1}{2\pi}\int_0^{2\pi}\sum_{n=0}^{\infty}c_n r^n e^{int}\sum_{m=0}^{\infty}\bar{c}_m r^m e^{-imt}\,dt \\
&= \sum_{n=0}^{\infty}\sum_{m=0}^{\infty}\frac{1}{2\pi}\int_0^{2\pi}c_n\bar{c}_m r^{n+m}e^{i(n-m)t}\,dt \\
&= \sum_{n=0}^{\infty}|c_n|^2 r^{2n}
\end{aligned}
$$

Here, we note that

$$
\int_0^{2\pi} e^{i(n-m)t} = \begin{cases} 2\pi & n=m \\ 0 & n\neq m \end{cases}
$$

□

Corollary. (Liouville theorem) All bounded entire functions are constant. Using this theorem, we can prove the fundamental theorem of algebra.

Proof. If f is entire such that $|f(z)| \leq M$ for all $z \in \boldsymbol{C}$ then

$$
\sum_{n=0}^{\infty}|c_n|^2 r^{2n} \leq M^2
$$

for all $r > 0$ which means that $c_1 = c_2 = \ldots = 0$. □

Theorem. (Identity theorem) Let D be a domain (open and connected) and $\{z_n\}$ be a sequence of distinct points in D such that $z_n \to z_0 \in D$. If f is analytic in D with $f(z_n) = 0$ for all $n \in \boldsymbol{N}$, then $f(z) = 0$ for all $z \in D$.

Theorem. (Maximum modulus theorem) Let f be analytic in a domain D. Then, $|f|$ can not have a local maximum in D unless it is constant.

Proof. If

$$
|f(z_0+re^{it}) \leq |f(z_0)|
$$

for all $t \in [0, 2\pi]$, then

$$
\sum_{n=0}^{\infty}|c_n|^2 r^{2n} \leq |f(0)|^2 = |c_0|^2
$$

which means that $c_1 = \ldots = c_n = 0$. Then we can apply the identity theorem. □

Theorem. (Morera's theorem) Morera's theorem can be thought as Cauchy theorem's inverse. If f is continuous in a domain D such that

$$\int_C f(z)\,dz = 0$$

for all closed contours C in D, then f is analytic in D (note that this also means that the integral is path independent).

Proof. The key point is that for a given $a \in D$, $F(z) = \int_a^z f(\xi)\,d\xi$ is well defined. Take any $z_0 \in D$. We'll show that $F'(z_0) = f(z_0)$, which means that $f = F'$ is analytic. Take any $\varepsilon > 0$. We choose $\delta > 0$ so that

(i) $D(z_0, \delta) \subset D$

(ii) If $|z - z_0| < \delta$ then $|f(z) - f(z_0)| < \varepsilon$

We need to prove that if $0 < |z - z_0| < \delta$ then

$$\left|\frac{F(z) - F(z_0)}{z - z_0} - f(z_0)\right| < \varepsilon$$

Take the line segment path $[z_0, z]$ from z_0 to z. Then

$$F(z) - F(z_0) = \int_{[z_0,z]} f(\xi)\,d\xi$$

and

$$\frac{F(z) - F(z_0)}{z - z_0} - f(z_0) = \frac{1}{z - z_0}\int_{[z_0,z]} [f(\xi) - f(z_0)]\,d\xi$$

Notice that now, due to (ii), the term inside the square brackets can be reduced to less than ε since $|\xi - z_0| < \delta$. Therefore

$$\left|\frac{F(z) - F(z_0)}{z - z_0} - f(z_0)\right| < \frac{1}{|z - z_0|}|z - z_0|\varepsilon = \varepsilon$$

□

Theorem. Let $\{f_n\}$ be a sequence of analytic functions in a domain D such that $f_n \to f$ uniformly on every compact sbuset of D. Then f is analytic on D.

Example. On $\boldsymbol{R}$, consider

$$f_n(x) = \sum_{k=1}^{n} \frac{\cos 3^k x}{2^k} \to f(x) = \sum_{n=1}^{\infty} \frac{\cos 3^n x}{2^n}$$

where the convergence is uniform and the right-hand side is nowhere differentiable.

Proof. First, f is continuous on D since convergence is uniform. Let C be any closed contour in D. Then

$$\int_C f(z)\,dz = \lim_{n\to\infty} \int_C f_n(z)\,dz = 0$$

By Morera's theorem, the proof is complete. $\square$

Lecture 3 (September 9th)

Theorem. Let f be analytic on a simply connected domain D such that $f(z) \neq 0$ for any $z \in D$. Then,

(i) There is g analytic on D such that $e^g = f$ on D

(ii) There is h analytic on D such that $h^2 = f$ on D (this implies the Riemann mapping theorem)

Proof. (1) By Cauchy's theorem, the $\int_C f(z)\,dz = 0$ for every closed contour C in D. Thus for a given $z_0 \in D$, the integral

$$\int_{z_0}^{z} f(\xi)\,d\xi$$

is independent of path. Since f has no zero in D, f'/f is analytic on D. We define g on D by

$$g(z) = \int_{z_0}^{z} \frac{f'(\xi)}{f(\xi)}\,d\xi + \operatorname{Log} f(z_0)$$

Then we'll show that $f(z) = e^{g(z)}$ on D. Observe that

$$\begin{aligned}
(fe^{-g})' =& f'e^{-g} + f(e^{-g})' \\
=& f'e^{-g} + fe^{-g}\Big(-\frac{f'}{f}\Big) \\
=& f'e^{-g} - f'e^{-g} = 0
\end{aligned}$$

on D and $fe^{-g} = c$ a constant. If we define $h(z) = f(z)e^{-g(z)}$ then

$$\begin{aligned}
h(z_0) =& f(z_0)e^{-g(z_0)} \\
=& f(z_0)e^{-\operatorname{Log} f(z_0)} \\
=& f(z_0)[f(z_0)]^{-1} = 1
\end{aligned}$$

(2) By (1), there exists an analytic g such that $f = e^g$ on D. If we define $h = e^{g/2}$, then h is analytic on D with $h^2 = f$. $\square$

Theorem. (Riemann mapping theorem) If D is a simply connected domain in $\boldsymbol{C}$ ($D \neq \boldsymbol{C}$), then there is a one-to-one analytic function f from D onto $U = \{z \in \boldsymbol{C} \,|\, |z| < 1\}$ (the unit disk).

Proof. The proof consists of three steps. Let Σ be a set of all 1-1 analytic functions from D to U.

$$\Sigma = \{f : D \to U \,/\, f \text{ is } 1-1 \text{ analytic}\}$$

We first prove that

(i) $\Sigma = \varnothing$

(ii) That if $\psi \in \Sigma$ satisfies $\psi(D) \neq U$, then for every $z_0 \in D$ there is $\psi_0 \in \Sigma$ such that $|\psi_0'(z_0)| > |\psi'(z_0)|$,

(iii) For a given $z_0 \in D$ if $\eta_0 = \sup\{|\psi'(z_0)| \,|\, \psi \in \Sigma\}$ then there is $f \in \Sigma$ such that $|f'(z_0)| = \eta_0$.

□

Theorem. (Exhastion of D by compact subsets) Let $D \subset \boldsymbol{C}$ be open. Then there is a sequence of compact subsets K_n of D such that

(i)
$$K_1 \subset K_2 \subset \ldots \subset K_n \subset D$$

(ii)
$$\bigcup_{n=1}^{\infty} K_n = D$$

(iii) If $K \subset D$ is compact then $K \subset K_n$ for some $n \in \boldsymbol{N}$

(iv) There is a sequence $\{\delta_n\}$ with $\delta_n > 0$ such that

$$\bar{D}(z, 2\delta_n) \subset K_{n+1}$$

for each $z \in K_n$

Proof. Define

$$K_n = \Big\{z \in \boldsymbol{C} \,|\, |z| \leq n\Big\} \cap \Big\{z \in \boldsymbol{C} \,|\, d(z, D^c) \geq \frac{1}{n}\Big\}$$

$$G_n = \Big\{|z| < n\Big\} \cap \Big\{d(z, D^c) > \frac{1}{n}\Big\}$$

and $G_n = D$ by the Archimeidan property. □

LECTURE 4 (SEPTEMBER 16TH)

Theorem. Let D be a domain in $\boldsymbol{C}$. There is a sequence $\{K_n\}$ of compact subsets of D and $\{\delta_n\}$ ($\delta_n > 0$) such that

(i)
$$\bigcup_n^{\infty} K_n = D$$

(ii) If $K \subset D$ is compact then $K \subset K_n$ for some n

(iii) $\bar{D}(z, 2\delta_n) \subset K_{n+1}$ for every $z \in K_n$, $n = 1, 2, \ldots$

Note that $z \notin F$ and F is closed. This implies that $d(z, F) = \min\{|z - w| \,|\, w \in F\} > 0$.

Proof. Put
$$K_n = \left\{z \leq n\right\} \cap \left\{d(z, D^C) \geq \frac{1}{n}\right\}$$
□

Example. Consider
$$f_n(x) = \frac{\sin(nx)}{\sqrt{n}} \to 0$$
which converges uniformly on $\boldsymbol{R}$. However,
$$f'(0) = \sqrt{n} \to \infty$$
as $n \to \infty$.

Theorem. If $\{f_n\}$ is a sequence of analytic functions on a domain D such that $f_n \to f$ uniformly on every compact subset of D, then

(i) f is analytic on D (by Morera's theorem)

(ii) $f_n' \to f'$ uniformly on every compact subset of D

Proof. Take any compact subset K of D. Then $K \subset K_n$ for some n, and $\bar{D}(z, \delta_n) \subset K_{n+1}$ for all $z \in K$. So, there exists $r > 0$ and K', a compact subset of D, such that $\bar{D}(z, r) \subset K'$ for all $z \in K$. Here, applying Cauchy's integral formula,
$$f(z) = \frac{1}{2\pi i} \int_{|\xi - z| = r} \frac{f(\xi)}{(\xi - z)^2} \, d\xi$$
For every $z \in K$, we have
$$f'(z) = \frac{1}{2\pi i} \int_{|\xi - z| = r} \frac{f(\xi)}{(\xi - z)^2} \, d\xi$$

which implies that

$$|f'(z)| \leq \frac{1}{2\pi} \frac{||f||_{K'}}{r^2}(2\pi r) = \frac{||f||_{K'}}{r}$$

with $||f||_K = \max_{z \in K} |f(z)|$. Therefore, for every $z \in K$,

$$|f_n'(z) - f'(z)| \leq \frac{||f_n - f||_{K'}}{r}$$

wihch means $f_n' \to f'$ uniformly on K. $\square$

Example. (Linear mapping) Consider the linear mapping

$$w = az + b$$

This mapping we see that, by expressing $a = re^{i\theta}$, the map stretches, rotates, and translates the complex plane.

Definition. (Linear functional transformation) For $ad - bc \neq 0$,

$$w = T(z) = \frac{az + b}{cz + d}$$

This can be written as

$$w = T(z) = \frac{a}{c} + \frac{bc - ad}{c} \cdot \frac{1}{cz + d}$$

Additionally, the map is 1-1 from the fact that the following inverse map exists:

$$z = \frac{dw - b}{-cw + a}$$

Thus telling us that the map can be seen as a combination of translation, dilation ($|a| > 1$) or contraction ($|a| \leq 1$), rotation, and inversion.

$$z \to \frac{1}{z}$$

This is the tricky one!

Remark. (Inversion map $w = 1/z$) The equation (locus) of all lines and all circles in $\boldsymbol{R}^2$ is

$$a(x^2 + y^2) + bx + cy + d = 0$$

where $b^2 + c^2 > 4ad$ and $a, b, c, d \in \boldsymbol{R}$. In the complex plane, for $x^2 + y^2 = |z|^2$, $x = (z + \bar{z})/2$ and $y = (z - \bar{z})/2i$ so that

$$a|z|^2 + \frac{1}{2}(b - ic)z + \frac{1}{2}(b + ic)\bar{z} + d = 0$$

and again $b^2 + c^2 > 4ad$. This implies that

$$a|z|^2 + \beta z + \bar{\beta}\bar{z} + d = 0$$

where $a, d \in \boldsymbol{R}$ and $\beta \in \boldsymbol{C}$ with $|\beta|^2 > ad$. Taking the inversion map,

$$a\frac{1}{|w|^2} + \beta\frac{1}{w} + \bar{\beta}\frac{1}{\bar{w}} + d|w|^2 = 0$$

and

$$a + \beta\bar{w} + \bar{\beta}w + d = 0$$

by multiplying $|w|^2 = w\bar{w}$ ($\beta\bar{\beta} > ad$). We therefore see that the inversion map takes the set of all lines and circles to again the set of all lines and circles.

Definition. (S^2) S^2 is defined as the boundary of a unit ball in $\boldsymbol{R}^3$. We see this as a one-point compactification (Alexandroff compactification) of $\boldsymbol{C}$, and is seen as $\boldsymbol{C} \cup \{\infty\}$ where $\{\infty\}$ is called the north pole.

Lemma. The linear functional transformation which takes $\{a, b, c\}$ to $\{0, 1, \infty\}$ is uniquely determined by

$$T(z) = \frac{(z-a)(b-c)}{(z-c)(b-a)}$$